

Meriken Park & Kitano, Kobe

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



メリケンパーク・北野（神戸市中央区）

Study area	2.2 km × 1.4 km around 34.6900° N, 135.1900° E
Oldest aerial photograph	1945–1950
Aerial series in this brief	1945–1950, 1961–1969, 1974–1978, 1979–1983, 1984–1986, 2021
Main natural landform	Alluvial fan (26%)
Former river channels	none mapped
Filled, reclaimed or drained land	29%
Former sea, river or pond	none mapped
Meiji-era paddies, marsh and flats	rice paddy 2.9%
Disaster memorials within 3 km	10 (earthquake 6, flood 4, landslide or debris flow 4)

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

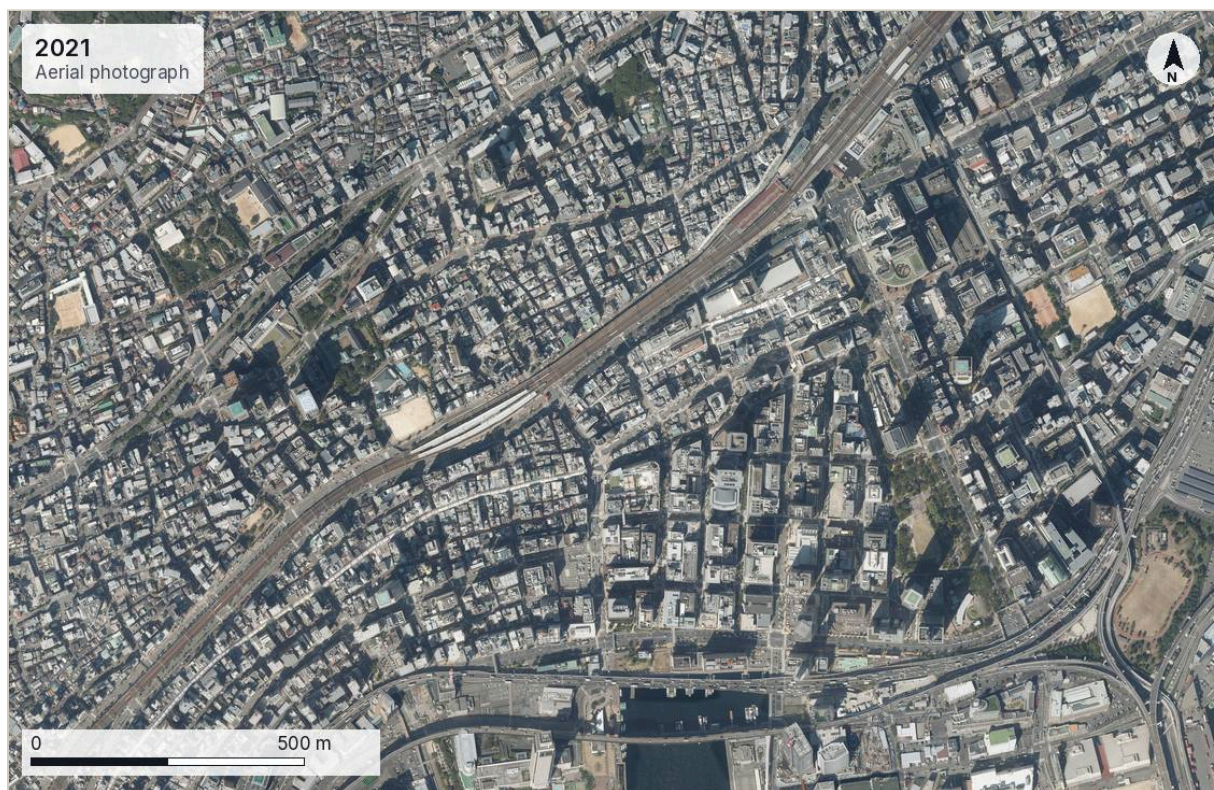
What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, 1945–1950 and 2021



1945–1950 aerial photograph (US military photography) — GSI Tiles *ort_USA10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2021 annual orthophoto — GSI Tiles *nendophoto2021*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



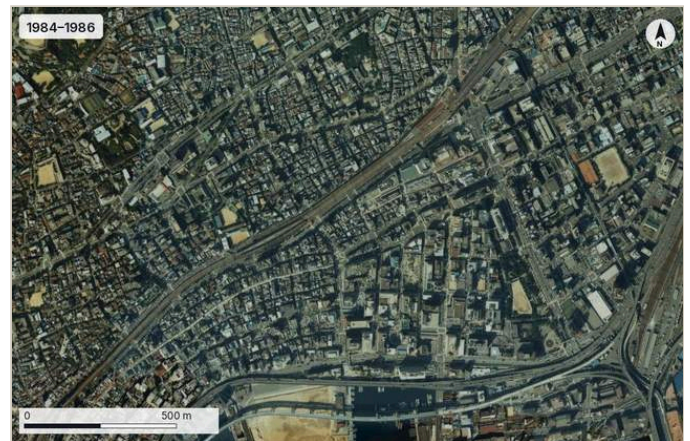
1961-1969 · GSI Tiles *ort_old10*, processed



1974-1978 · GSI Tiles *gazo1*, processed

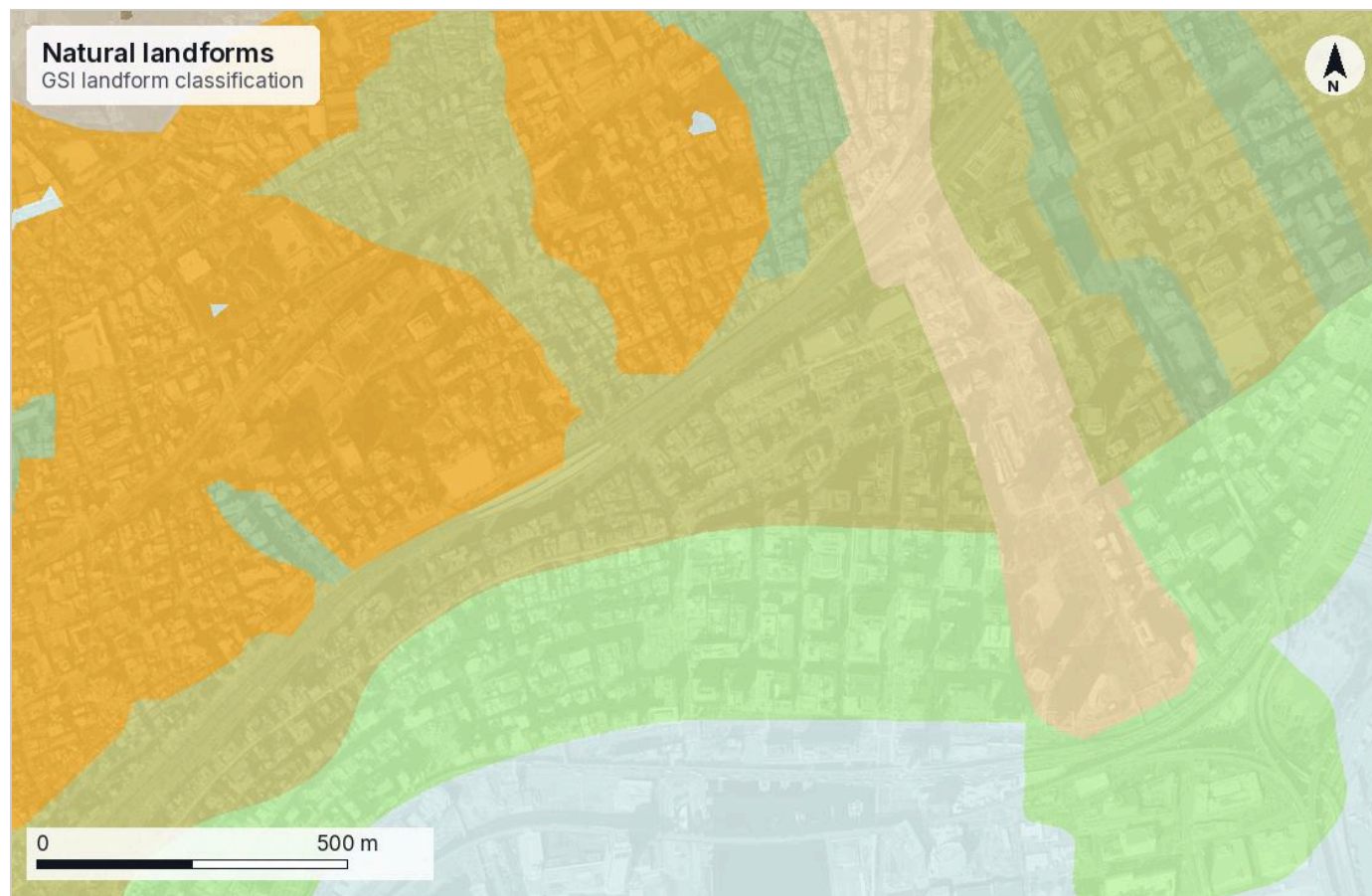


1979-1983 · GSI Tiles *gazo2*, processed



1984-1986 · GSI Tiles *gazo3*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.

 Alluvial fan 26%	 Plateaus and terraces 25%
 Floodplain and coastal plain 21%	 Water 11%
 Hollows and shallow valleys 7.4%	 Raised (perched) river 7.3%
 Mountain and hill slopes 1.9%	

What the main landforms mean

Alluvial fan (26%). Gently sloping ground spreading fan-like from a valley mouth, built from sediment dropped by floods. Floodwater from the mountains, and debris flows near the valley mouth. Relatively firm ground; a slight liquefaction tendency on the lower fan.

Plateaus and terraces (25%). Flat ground standing in steps above its surroundings, left behind as the land around it was eroded. River flooding is rarely a threat unless the terrace is only slightly above the river; watch the slopes at its edges. Firm ground: shaking is relatively weak and liquefaction is unlikely.

Floodplain and coastal plain (21%). Low, flat ground of little relief, built from sand and mud left by floods, or former seabed that emerged. River flooding. Ground is softer nearer the coast and shakes somewhat more; liquefaction tendency is strong on coastal plains and somewhat strong on floodplains. Storm surge near the coast.

Hollows and shallow valleys (7.4%). Slightly lower ground within terraces, fans or dunes, shaped by small streams or by sediment building up around it. Rainwater gathers here in downpours; the ground can be softer than its surroundings, with a strong liquefaction tendency among sandbars and dunes.

Raised (perched) river (7.3%). Stretches where a river runs higher than the land around it, and the slightly raised ground along them. If a perched river breaks its embankment, floodwater spreads over the surroundings very quickly.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.



GSI “Meiji-era lowland” data, traced from topographic maps of the 1880s–1900s; GSI warns that positions can be off by a considerable distance, especially in the Kanto and Kinki regions. Only low, wet or open ground was traced: towns, dry fields and forest of the period are left blank.

Man-made ground

-  Filled and reclaimed land 29%
-  Cut ground 13%

Filled and reclaimed land (29%). Ground raised with fill, land made by dumping soil into the sea, rivers or ponds, or valleys filled in. May flood if not raised enough; on hills, rain or earthquakes can make the fill fail; in lowlands the liquefaction tendency is very strong, especially on filled sea, lake or river beds.

Cut ground (13%). Ground made by cutting into slopes of mountains or terrace edges. Depending on the cut slope, heavy rain or earthquakes may bring it down; the ground itself is generally firm.

Meiji-era lowland

-  River, pond or sea 7.0%
-  Rice paddy 2.9%

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 6 km across.

Memorial	Disaster	Erected	Distance
1 Monument for Remembrance and Reconstruction, with the 1.17 Light of Hope 慰霊と復興のモニュメント・1.17希望の灯り	Great Hanshin-Awaji Earthquake (17 Jan 1995) · earthquake	2000	0.7 km
2 Port of Kobe Earthquake Memorial Park 神戸港震災メモリアルパーク	Great Hanshin-Awaji Earthquake (17 Jan 1995) · earthquake	1997	0.7 km
3 Monument to Kano-cho's recovery from the 1938 Hanshin flood 水害復興記念碑	Great Hanshin flood (5 Jul 1938) · flood / landslide or debris flow	c. 1940	0.9 km
4 Kobe Earthquake Reconstruction Memorial Park (Minatonomori Park) 神戸震災復興記念公園(みなとのもり公園)	Great Hanshin-Awaji Earthquake (17 Jan 1995) · earthquake	2010	1.0 km
5 “Flowing Stones” monument to the 1938 Hanshin flood 流石之碑	Great Hanshin flood (5 Jul 1938) · flood / landslide or debris flow	1939	1.4 km
6 Monument to the recovery from the 1938 Hanshin flood 水害復興記念	Great Hanshin flood (5 Jul 1938) · flood / landslide or debris flow	1938	2.2 km
7 Memorial tower to the victims of the 1938 Hanshin flood 慰霊塔	Great Hanshin flood (5 Jul 1938) · flood / landslide or debris flow	1941	2.3 km
8 Requiem monument to the victims of the 1995 Great Hanshin-Awaji Earthquake 鎮魂の碑	Great Hanshin-Awaji Earthquake (17 Jan 1995) · earthquake	2002	2.5 km
9 Memorial to those who died in the 1995 Great Hanshin-Awaji Earthquake 阪神淡路大震災物故者慰霊碑	Great Hanshin-Awaji Earthquake (17 Jan 1995) · earthquake	1995	2.7 km
10 Monument of Life (Inochi no Hi) for the 1995 Great Hanshin-Awaji Earthquake いのちの碑	Great Hanshin-Awaji Earthquake (17 Jan 1995) · earthquake	2004	2.8 km

What the memorials record

1. **Monument for Remembrance and Reconstruction, with the 1.17 Light of Hope.** The Great Hanshin-Awaji Earthquake of 17 January 1995 left 6,437 people dead or missing. The monument keeps the disaster in memory, passes on the story of the recovery, honours the victims, encourages citizens and speaks for the value of worldwide solidarity in recovering from major disasters.
2. **Port of Kobe Earthquake Memorial Park.** A 60 m stretch of Meriken Wharf, badly damaged in the Great Hanshin-Awaji Earthquake of 1995, is preserved as it was left. The park shows how the Port of Kobe was damaged and rebuilt, and hands down

the lessons of the earthquake, the importance of the port and the joint effort that restored it.

3. Monument to Kano-cho's recovery from the 1938 Hanshin flood. Days of summer rain peaked in a downpour on 5 July 1938, setting off debris flows across Mount Rokko and flooding rivers all at once. Torrents from the Nunobiki valley and nearby streams tore into Kano-cho, killing 12 people and damaging over 220 houses. The monument was moved here in March 2007 from 200 m to the north-east.

4. Kobe Earthquake Reconstruction Memorial Park (Minatonomori Park). The park was created as a symbol to nurture the spirit of the aid and volunteers that came from across Japan and the world after the Great Hanshin-Awaji Earthquake of 1995. Its guiding idea is to watch Kobe move from recovery to growth as its trees grow.

5. "Flowing Stones" monument to the 1938 Hanshin flood. The downpour of 5 July 1938 set off debris flows across Mount Rokko; the Uji and Kusudani rivers in Kusudani-cho overflowed, sweeping away about 150 houses and killing more than 10 people. Made of boulders from the flood, the monument records the damage and notes that, as everyone joined in the recovery, its name may be read "sasuga" ("truly admirable").

6. Monument to the recovery from the 1938 Hanshin flood. The downpour of 5 July 1938 set off debris flows across Mount Rokko, and rivers overflowed all at once. Arata-cho was swept by muddy torrents in an instant, with 125 victims and about 1,800 houses flooded.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_USA10* (1945–1950, US military photography), *ort_old10* (1961–1969), *gazo1* (1974–1978), *gazo2* (1979–1983), *gazo3* (1984–1986) and *nendophoto2021*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.2×1.4 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

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